

```
%DAY 1 (5 june 2026) Internship CoE-mathworks deeptech labs
%https://in.mathworks.com/help/matlab/getting-started-with-matlab.html?
s_tid=CRUX_lftnav
%name: HUJEFA
%reg no: 2023BECE07AED033
%2023-27 batch %
%VII sem
%Alliance university
```

Desktop Basics

```
% 1st part
clc;
clear all;
close all;
a=5;
b=6;
c=[1 4 6]
```

```
c = 1×3
     1     4     6
```

```
%2nd part
d=[1,2;2,2]
```

```
d = 2×2
     1     2
     2     2
```

```
a=sin(d)
```

```
a = 2×2
     0.8415     0.9093
     0.9093     0.9093
```

```
A=10;
t=[0,1,2,3,4,5,6,7,8,9,10]
```

```
t = 1×11
     0     1     2     3     4     5     6     7     8     9    10
```

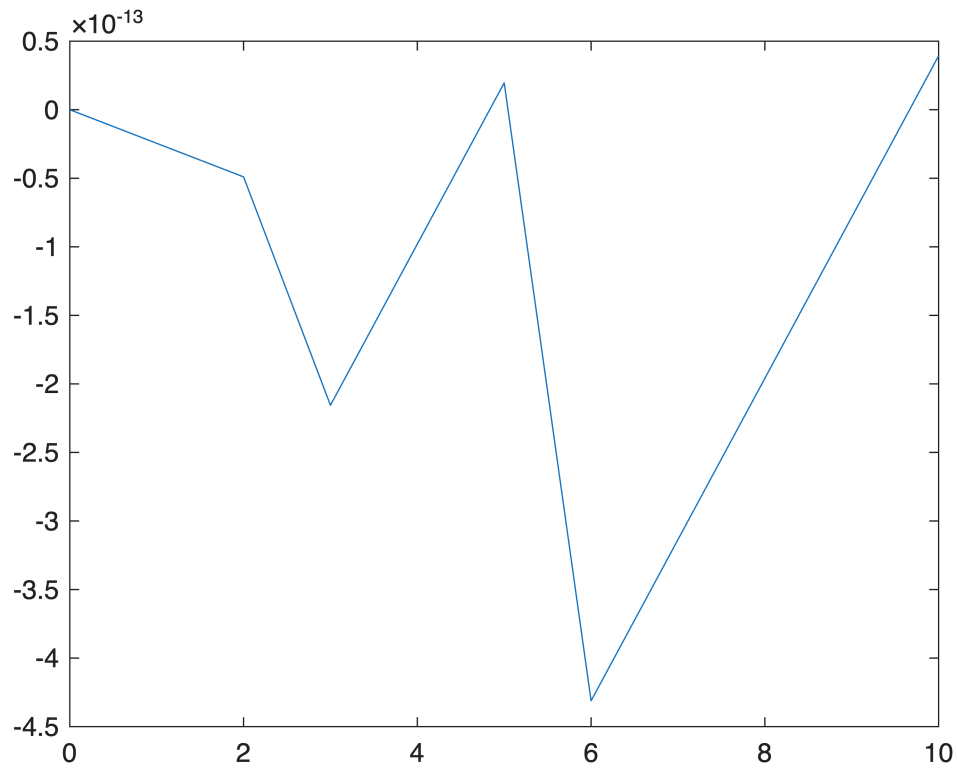
```
f=10
```

```
f =
10
```

```
ms=A*sin(2*pi*f*t)
```

```
ms = 1×11
10-12 ×
     0    -0.0245    -0.0490    -0.2156    -0.0980     0.0196    -0.4312    -0.3136 ...
```

```
plot(t,ms) %command for plotting
```



Matrices and Arrays

```
%array creation
```

```
a=[1 2 3 4]
```

```
a = 1x4
     1     2     3     4
```

```
%Matrix creation 3x3
```

```
b=[1 3 5;3 4 6;1 2 5]
```

```
b = 3x3
     1     3     5
     3     4     6
     1     2     5
```

```
c= [1 3 5
     2 4 6
     7 8 10]
```

```
c = 3x3
     1     3     5
```

2	4	6
7	8	10

```
%4x4 matrix
```

```
a=[1 2 3 4; 2 3 4 5; 1 2 3 4;7 8 9 0]
```

```
a = 4x4
```

1	2	3	4
2	3	4	5
1	2	3	4
7	8	9	0

```
%only zeros matrix (5x5)
```

```
z1=zeros(5,5)
```

```
z1 = 5x5
```

0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

```
%only ones matrix (5x5)
```

```
z=ones(5,5)
```

```
z = 5x5
```

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Matrix and Array Operations

```
z+10 %addition
```

```
ans = 5x5
```

11	11	11	11	11
11	11	11	11	11
11	11	11	11	11
11	11	11	11	11
11	11	11	11	11

```
z-10 %subtraction
```

```
ans = 5x5
```

-9	-9	-9	-9	-9
-9	-9	-9	-9	-9
-9	-9	-9	-9	-9
-9	-9	-9	-9	-9
-9	-9	-9	-9	-9

```
z*10 %multiplication
```

```
ans = 5×5
    10    10    10    10    10
    10    10    10    10    10
    10    10    10    10    10
    10    10    10    10    10
    10    10    10    10    10
```

```
z/10 %division
```

```
ans = 5×5
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000
```

```
%inverse function
```

```
p = a*inv(a)
```

```
Warning: Matrix is singular to working precision.
```

```
p = 4×4
    Inf    Inf    I
    Inf    Inf    I
    Inf    Inf    I
    NaN    NaN    N
```

```
p = a.*a
```

```
p = 4×4
     1     4     9    16
     4     9    16    25
     1     4     9    16
    49    64    81     0
```

```
a.^3 %power
```

```
ans = 4×4
     1     8    27    64
     8    27    64   125
     1     8    27    64
    343   512   729     0
```

```
a.^5
```

```
ans = 4×4
     1     32    243   1024
    32    243   1024   3125
     1     32    243   1024
   16807   32768   59049     0
```

%Concatenation

```
A = [a,a]
```

```
A = 4×8
```

1	2	3	4	1	2	3	4
2	3	4	5	2	3	4	5
1	2	3	4	1	2	3	4
7	8	9	0	7	8	9	0

```
A = [a; a]
```

```
A = 8×4
```

1	2	3	4
2	3	4	5
1	2	3	4
7	8	9	0
1	2	3	4
2	3	4	5
1	2	3	4
7	8	9	0

%complex numbers

```
sqrt(-1) %square root of -1
```

```
ans =
```

```
0.0000 + 1.0000i
```

```
c = [3+4i, 4+3j; -i, 10j] %matrix
```

```
c = 2×2 complex
```

3.0000 + 4.0000i	4.0000 + 3.0000i
0.0000 - 1.0000i	0.0000 +10.0000i

Text and Characters

%Text in String Arrays

```
t= "hello, world";
```

```
whos t %identity of t
```

Name	Size	Bytes	Class	Attributes
t	1x1	176	string	

```
f = 71;
```

```
c = (f-32)/1.8;
```

```
tempText = "Temperature is " + c + "C" %concatenation
```

```
tempText =
```

```
"Temperature is 21.6667C"
```

```
A = ["a", "bb", "ccc"; "dddd", "eeeeee", "fffffff"]
```

```
A = 2×3 string
"a"      "bb"      "ccc"
"dddd"   "eeeeee"  "fffffff"
```

```
strlength(A) %command for length of a string
```

```
ans = 2×3
     1     2     3
     4     6     7
```

Data in Character Arrays

```
h='huzefa p';
whos h %identity of h
```

Name	Size	Bytes	Class	Attributes
h	1x8	16	char	

```
h(5) %single character.
```

```
ans =
'f'
```

```
h2=[h 'hello'] %Concatenate character arrays with square brackets
```

```
h2 =
'huzefa phello'
```

Calling Functions

```
max(a) %max of matrix (a)
```

```
ans = 1×4
     7     8     9     5
```

```
a=[1 3 5];
max(a)
```

```
ans =
5
```

```
% maximum of (a)
```

```
b=[3 5 6];
union(a,b) %union of a and b
```

```
ans = 1×4
     1     3     5     6
```

```
max(b)
```

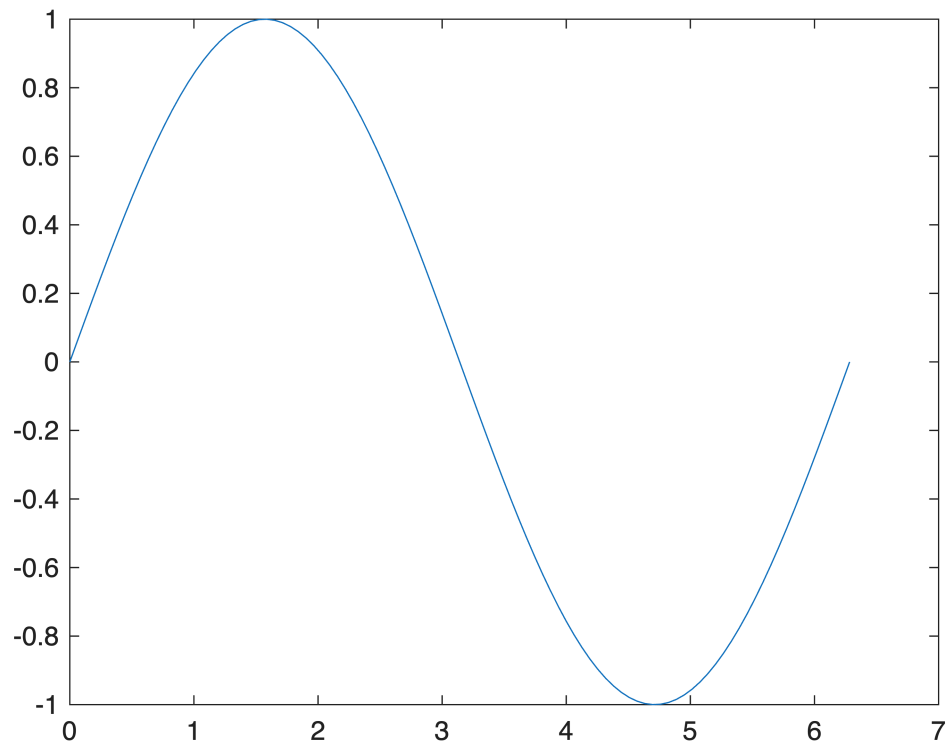
```
ans =  
6
```

```
disp("hello world") %display or print
```

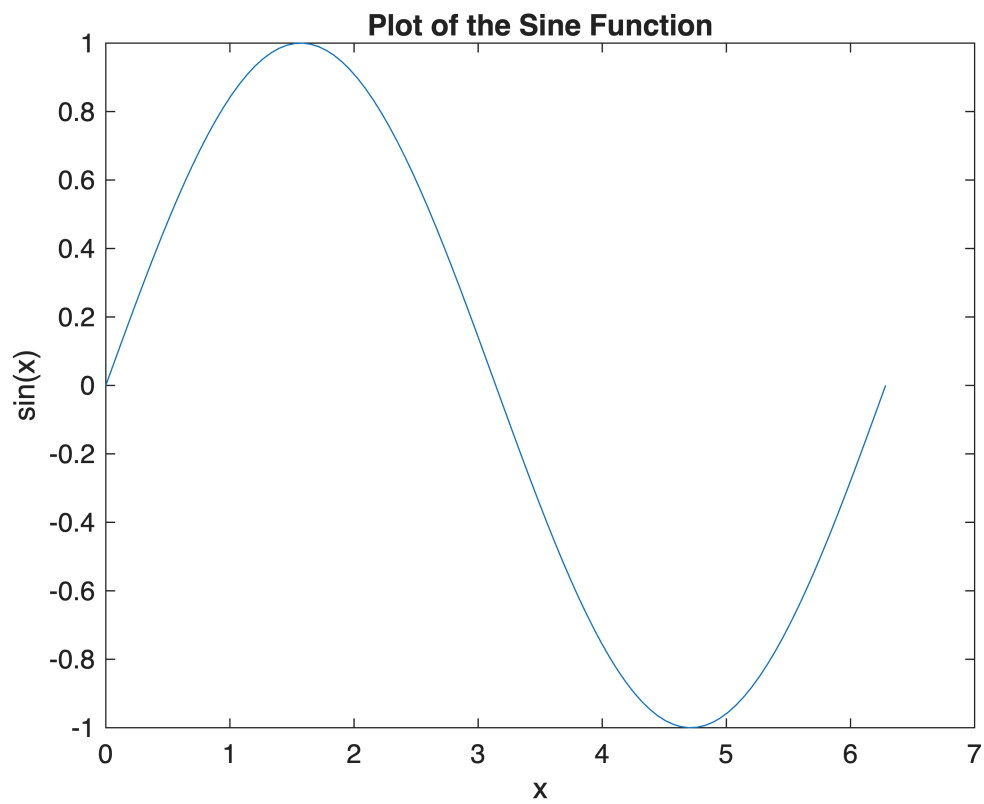
```
hello world
```

2-D and 3-D Plots

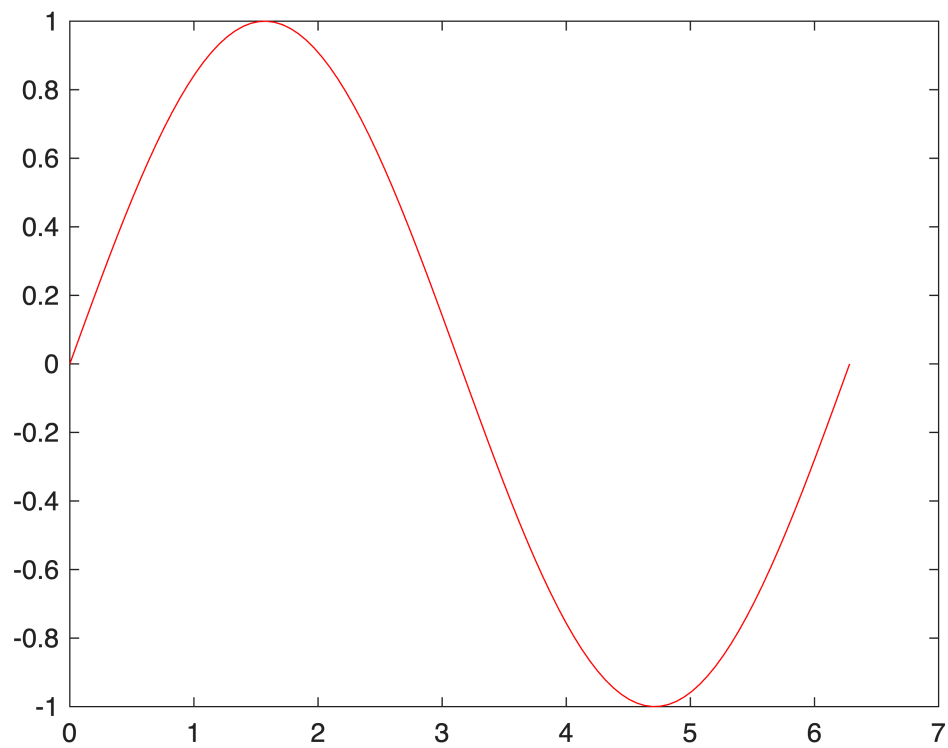
```
%Line Plots  
x=linspace(0,2*pi);  
y=sin(x);  
plot(x,y) %command of plotting
```



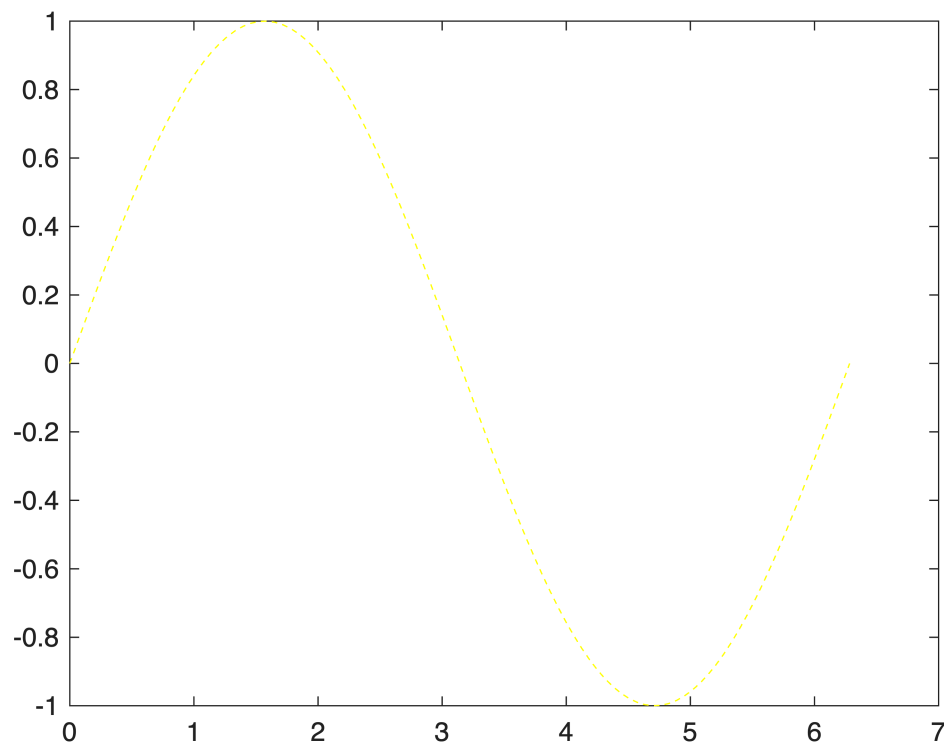
```
%giving title  
xlabel("x")  
ylabel("sin(x)")  
title("Plot of the Sine Function")
```



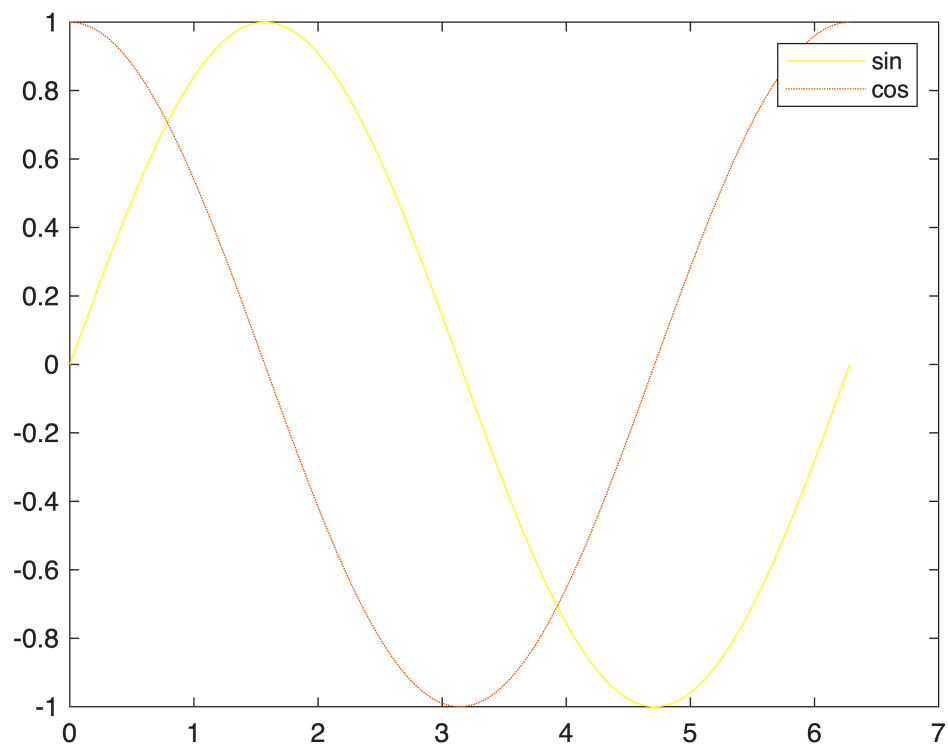
```
plot(x,y,"r") %command for colour (r=red)
```




```
plot(x,y,"y--") %use -- for dotted line %command for colour (y=yellow)
```

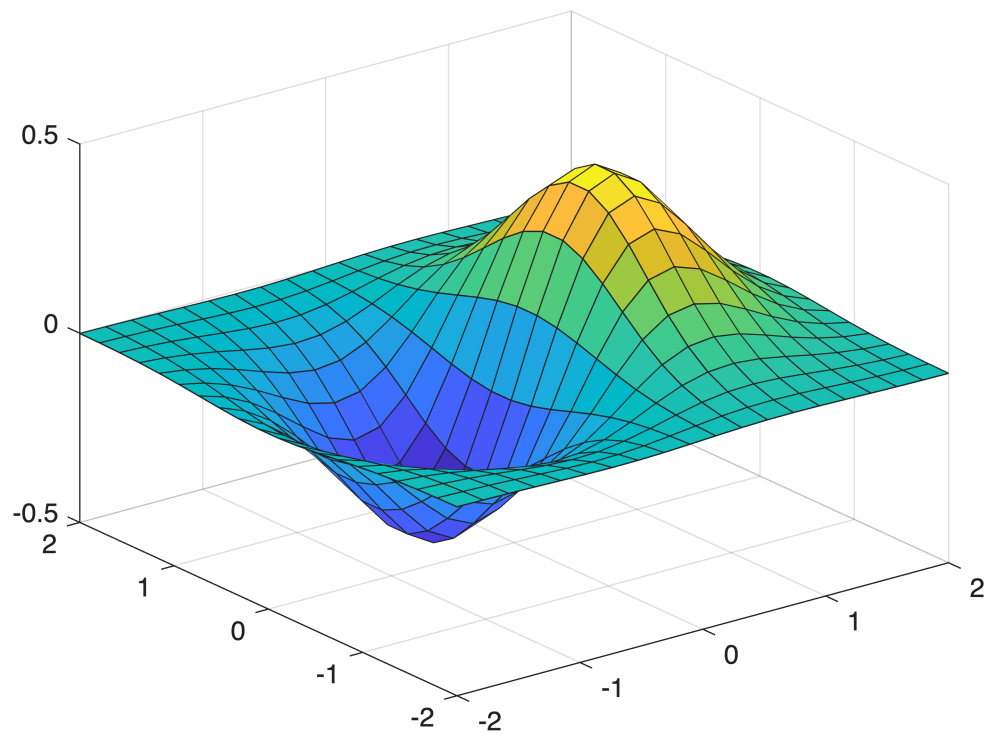


```
x = linspace(0,2*pi);  
y = sin(x);  
plot(x,y,"y") %"y" for colour (yellow)  
  
hold on  
  
y2 = cos(x);  
plot(x,y2,":") %use : for dotted lines  
legend("sin","cos")  
  
hold off
```

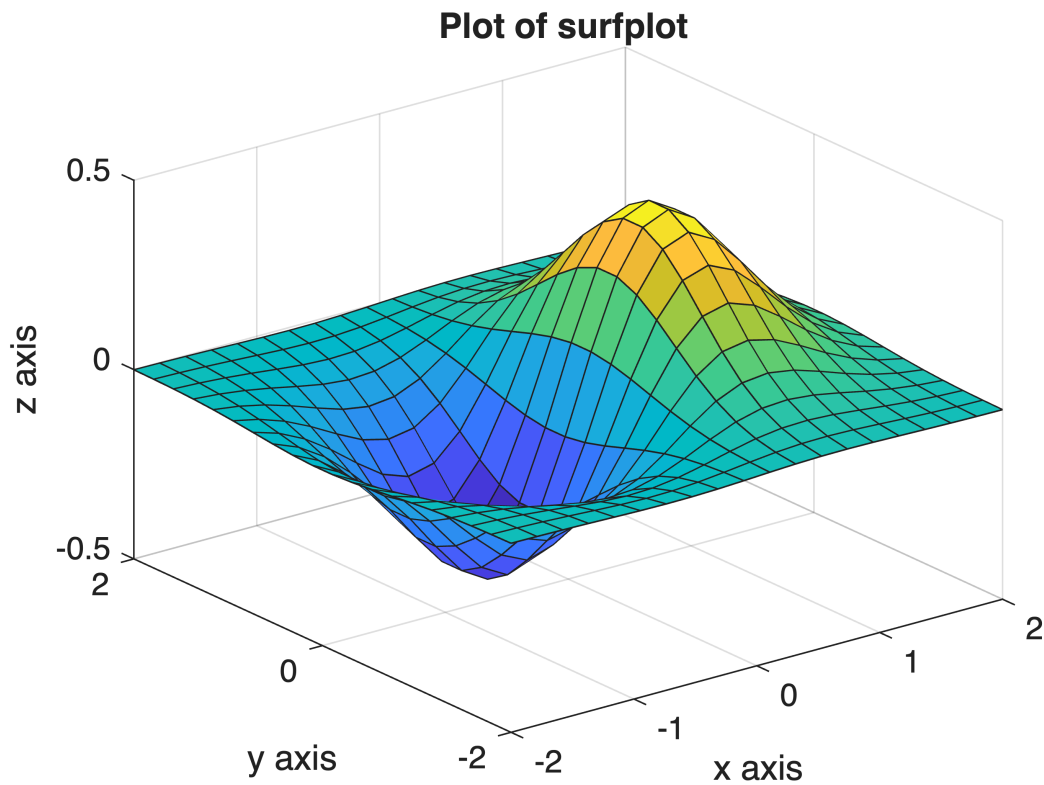


3-D Plots

```
x=linspace(-2,2,20);  
y=x';  
z=x.*exp(-x.^2-y.^2);  
surf(x,y,z)
```



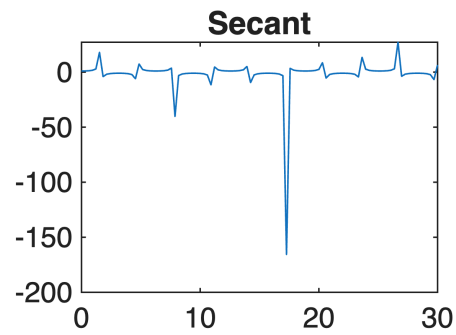
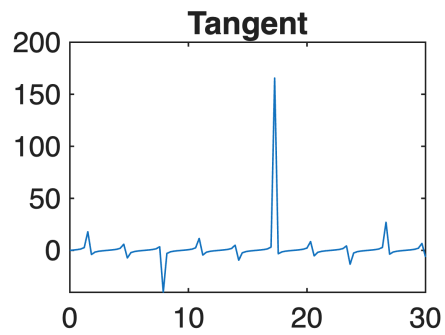
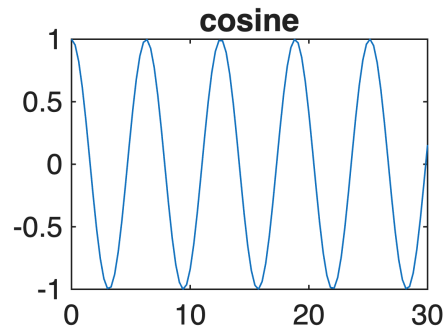
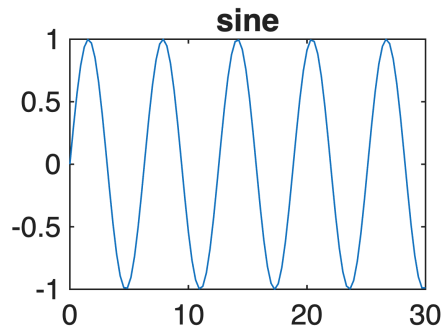
```
%giving title and labels  
xlabel("x axis")  
ylabel("y axis")  
zlabel("z axis")  
title("Plot of surfplot ")
```



Multiple Plots

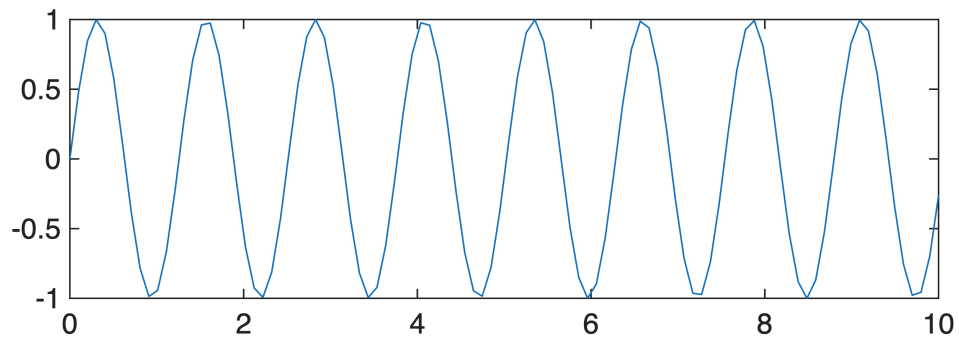
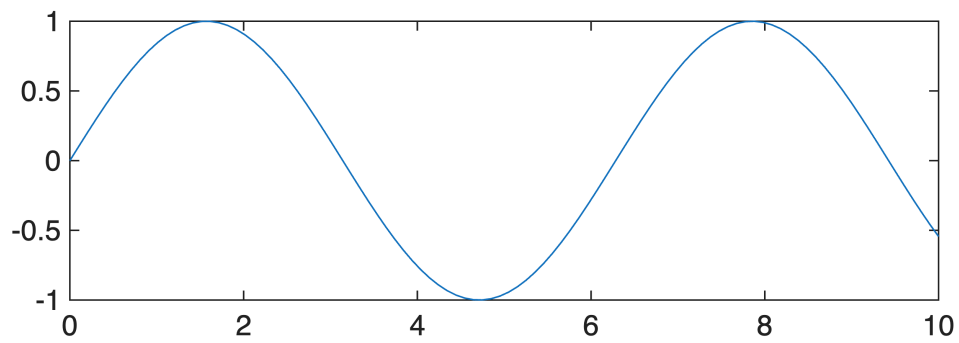
```
t = tiledlayout(2,2);  
title(t,"Trigonometric functions")  
x=linspace(0,30);  
  
nexttile  
plot(x,sin(x))  
title("sine")  
  
nexttile  
plot(x,cos(x))  
title("cosine")  
  
nexttile  
plot(x,tan(x))  
title("Tangent")  
  
nexttile  
plot(x,sec(x))  
title("Secant")
```

Trigonometric functions



Subplotting

```
subplot(2,1,1);  
x=linspace(0,10);  
y1=sin(x);  
plot(x,y1)  
  
subplot(2,1,2);  
y2=sin(5*x);  
plot(x,y2)
```



```
%stem function
```

```
x=1:10
```

```
x = 1×10
```

```
1    2    3    4    5    6    7    8    9   10
```

```
y=1:10
```

```
y = 1×10
```

```
1    2    3    4    5    6    7    8    9   10
```

```
stem(x,y)
```

